Optimal Tax Problem (with another open question)

Here's the question I left you to think about for Thursday. Let d be a demand correspondence from a LNS, continuous, strictly quasiconcave utility (and so it is indeed a demand *function*, and it satisfies budget balance and WARP). Let $t \in \mathbb{R}^L_+$. (Hmm, let's remember to look in the bottom of the IKEA box for this last assumption.) And let $x^t = d(p+t, w)$, $T = t \cdot x^t$, and $x^\ell = d(p, w-T)$.

There are two open questions: Can you say whether $x^t \mathcal{R} x^\ell$ or $x^\ell \mathcal{R} x^t$ or both or neither? And what can you say about the choices x^t vs x^ℓ? Again if you get stuck I suggest you use the special case $t = (t_1, 0) > 0$ and the associated picture for inspiration.

$\star$ If you make some progress, here is another 'open question' to think about. Is it possible that $x^t = x^\ell$? If not, prove that the equality is impossible. If so give as many ways this can happen as you can. (Hint: it's either 0 or 3.)

You might want to read MWG, chapter 2, on the Compensated Law of Demand (CLD). The first question is: what in the world does it have to do with the optimal tax problem?